

Pre Survey Checklist for Boilers

🔥 Introduction to Boiler Types: Steam vs. Hydronic Systems

Boiler systems are essential for heating and process operations across commercial, industrial, and institutional facilities. The two most common types — Steam Boilers and Hydronic Boilers — serve similar purposes but operate through distinct methods and safety requirements.

Steam Boilers generate and distribute steam at high temperatures and pressures, making them ideal for heavy-duty applications such as manufacturing, sterilization, and process heating. These systems require careful pressure control, condensate return management, and frequent blowdown to prevent scale and maintain safety compliance.

Hydronic Boilers, on the other hand, circulate hot water through a closed-loop system, offering efficient temperature control, zoning flexibility, and quiet operation. Common in commercial buildings, schools, and residential systems, they rely on flow switches, aquastats, and expansion tanks to maintain pressure, flow, and heat distribution safely. Understanding the fundamental differences between these systems is critical to ensuring proper installation, optimization, and maintenance — all while aligning with OSHA, ASME, and NFPA safety standards.